

Match-the-Graph: what the graph says

Walk a position–time graph with the Go!Motion sensor — then say what would prove it wrong.

Name _____ Date _____ Group _____

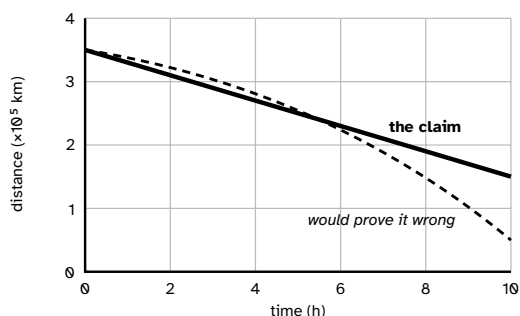
TODAY'S TARGET I can capture a position–time graph with a Vernier motion detector and explain what its slope tells me about velocity.	ASTEROID THREAD · 2026-XJ Radar hands us position over time. A tracking graph is not a picture — it is a claim about how the object moved. Claims can be wrong. Today you learn to test one with your own body.
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LAB SETUP Driving question: What does a position–time graph assert about the world — and what would prove it wrong? Procedure 1. Open Graphical Analysis , Go!Motion on USB. Axes: time 0–10 s, position 0–4 m . Sensor sits at 0 m. 2. Before anyone walks: each person writes the graph's claim and one walk that would prove it wrong . 3. Walk it. Sketch what the screen showed. (Graphs 4–6: also sketch the v–t graph you expect first.) 4. Disagree, adjust, walk again. Rotate the walker every graph.	Equipment (per group of 3) • 1 Vernier Go!Motion + USB cable • 1 laptop with Graphical Analysis • 4 m clear lane; tape marks at 1, 2, 3 m Safety: walk, don't run. Face where you are going; partners read the screen. Stay inside the lane. <i>How this lab serves the year's question: if we can't say what would prove a graph wrong, we can't trust it — with a walker or an asteroid.</i>
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★ Sign convention: away from the sensor is +, toward is –. A p–t slope is a velocity; a curve means the velocity is changing.
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WORKED EXAMPLE — WHAT A GRAPH SAYS, AND WHAT WOULD PROVE IT WRONG

Radar: 2026-XJ distance from Earth



What the solid line says: 2026-XJ is getting closer to Earth at one steady rate — 2.0×10^5 km every 10 h. Never faster, never slower, no turns. The slope is -0.2×10^5 km/h everywhere on the line. That is the story the graph tells.

What would make the story wrong: any speeding up. If Earth's gravity were already pulling harder, the line would bend steeper as time passes (dashed). So a bend is the thing to look for. If we see one, the straight-line story is wrong.

Everyday version: an elevator's floor-vs-time graph that is a straight rising line says it never stopped. One passenger who got off on the 4th floor makes that story wrong — there would have to be a flat step.

The line is exact. The data is not. The drawn line says exactly -0.2×10^5 km/h. Real radar readings do not land exactly on it — each one is off by a little. So a point slightly off the line does *not* mean the story is wrong. Points far off, or drifting off in one direction, do. Extend the line to next June and that small wobble becomes a large spread in where the asteroid could be. Keep that in mind.

Pattern: say what the slope says stays the same. Say what shape would break it. Then ask how much wobble in the points you would accept.

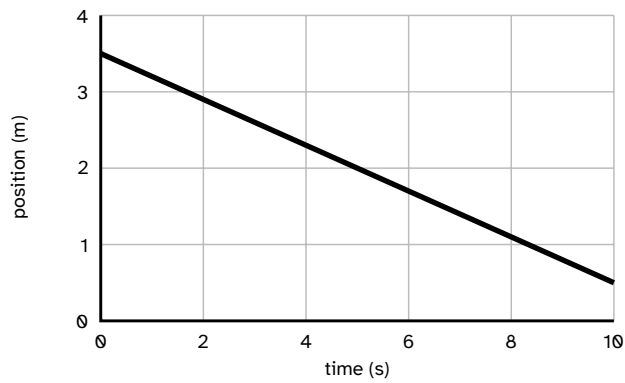
Part A • Read it, break it, walk it

For each target: write what the graph **claims** (start, direction, speed, changes). For Graphs 1–3, name one walk that would **prove it wrong**. For Graphs 4–6, sketch the **v–t** graph instead and say how you got it from the slopes. Then walk it and record what the screen showed.

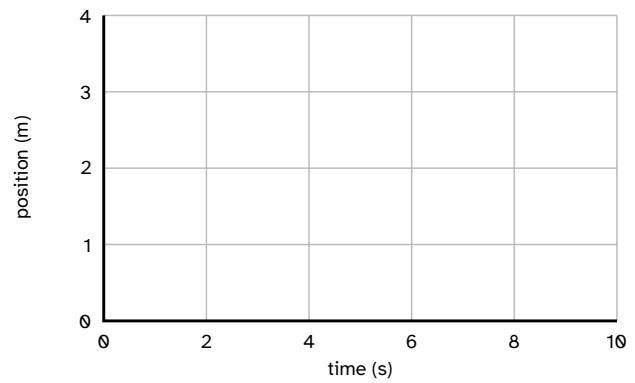
GRAPH 1

walker: _____

Target · position-time



What the screen showed



This graph **claims**: _____

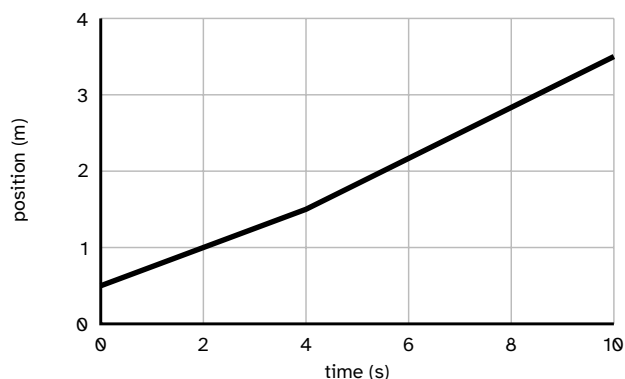
A walk that would **prove it wrong**: _____

Slope of straight part(s): _____ m/s Matched? [**yes / almost / no**] Change: _____

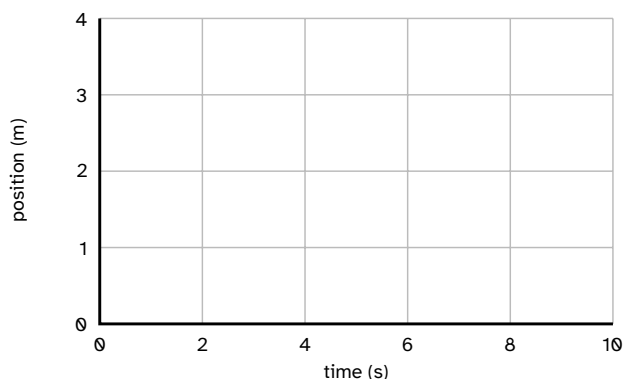
GRAPH 2

walker: _____

Target · position-time



What the screen showed



This graph **claims**: _____

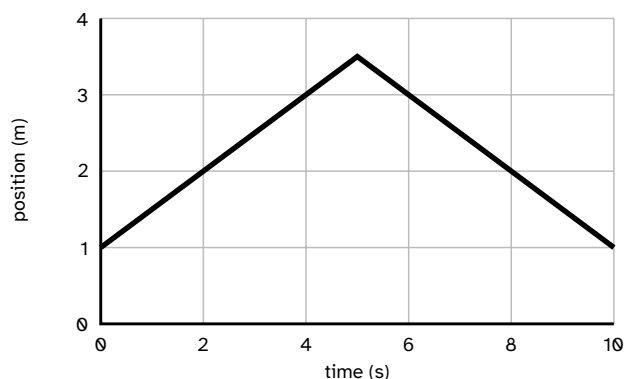
A walk that would **prove it wrong**: _____

Slopes: 0–4 s _____ m/s · 4–10 s _____ m/s Matched? [**yes / almost / no**] Change: _____

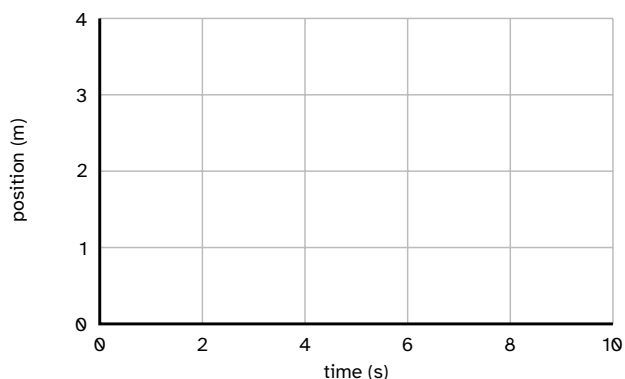
GRAPH 3

walker: _____

Target · position-time



What the screen showed



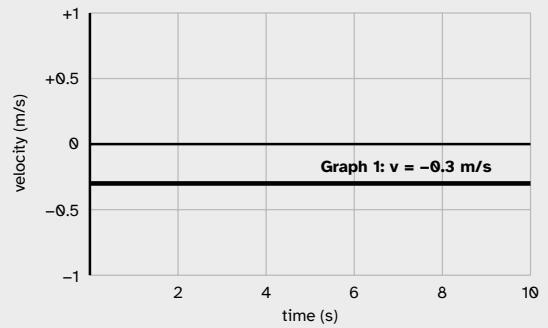
This graph **claims**: _____

A walk that would **prove it wrong**: _____

Slopes: 0–5 s _____ m/s · 5–10 s _____ m/s At $t = 5$ s the velocity is _____ Matched? [**yes / almost / no**]

New for Graphs 4–6: sketch the velocity–time graph

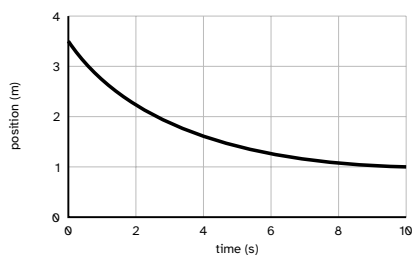
too. Read each straight segment's slope from the p–t graph; that number is the height of the v–t graph on that interval. Flat p–t $\rightarrow v = 0$. Rising $\rightarrow$ above zero. Falling $\rightarrow$ below zero. Curved p–t $\rightarrow$ a sloped v–t. Check yourself: Graph 1 (slope -0.3 m/s the whole way) has the v–t graph at right.



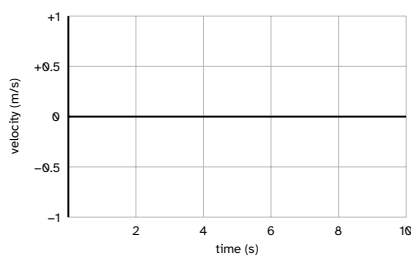
GRAPH 4

walker: _____

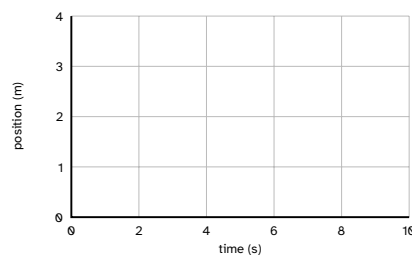
Target · position-time



Predicted velocity-time



What the screen showed



This graph **claims**: _____

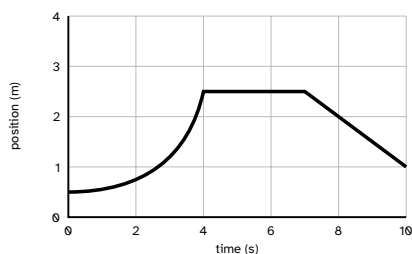
How my **v-t sketch** follows from the slopes:

No straight part. The slope is [**getting steeper / getting flatter**], so the speed is [**increasing / decreasing**]. Matched? [**yes / almost / no**]

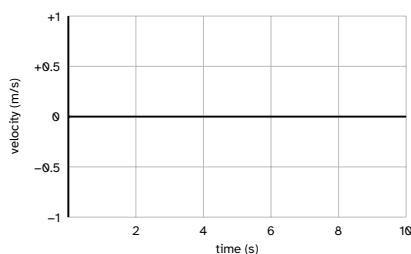
GRAPH 5

walker: _____

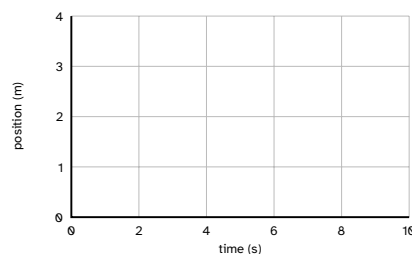
Target · position-time



Predicted velocity-time



What the screen showed



This graph **claims**: _____

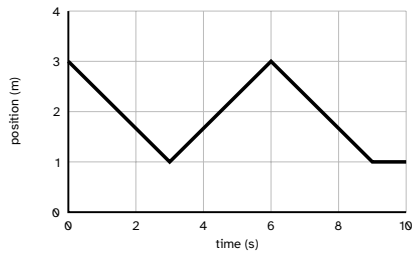
How my **v-t sketch** follows from the slopes:

Slope 4-7 s: _____ m/s · Slope 7-10 s: _____ m/s Matched? [**yes / almost / no**] Change:

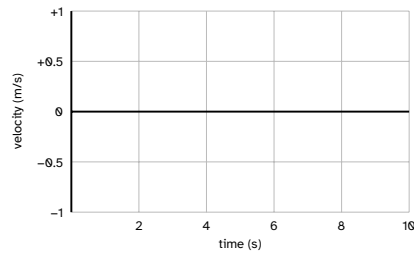
GRAPH 6

walker: _____

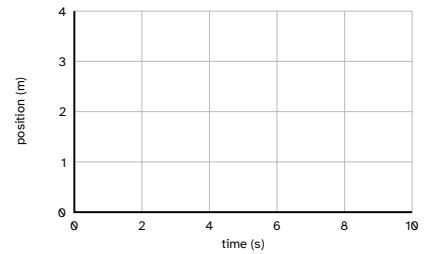
Target · position-time



Predicted velocity-time



What the screen showed



This graph **claims**: _____

How my **v-t sketch** follows from the slopes:

Slopes: 0–3 s _____ · 3–6 s _____ · 6–9 s _____ · 9–10 s _____ m/s Which was hardest to walk, and why?

Part B • Slope = velocity, with its sign

One model, then you. Every straight segment above gets a number in Part A; here you show the method once in full.

WORKED EXAMPLE — GRAPH 1		
Graph 1 is one straight line from (0 s, 3.5 m) to (10 s, 0.5 m). Find the velocity, with sign.		
GIVEN	$t_1 = 0 \text{ s}, x_1 = 3.5 \text{ m} \cdot t_2 = 10 \text{ s}, x_2 = 0.5 \text{ m}$	<i>Two points on the line.</i>
EQUATION	$v = \Delta x / \Delta t = (x_2 - x_1) / (t_2 - t_1)$	<i>Order matters: final minus initial.</i>
WORK	$v = (0.5 \text{ m} - 3.5 \text{ m}) / (10 \text{ s} - 0 \text{ s}) = -3.0 \text{ m} / 10 \text{ s}$	<i>The minus sign comes out of the subtraction — you do not add it by hand.</i>
ANSWER	−0.3 m/s — The walker moved toward the sensor at 0.3 m every second. The negative sign is the direction.	<i>Speed 0.3 m/s; velocity −0.3 m/s.</i>

YOUR TURN — GRAPH 3, BOTH SEGMENTS		
GIVEN	0–5 s	5–10 s
EQUATION		
WORK		
ANSWER	$v = \underline{\hspace{2cm}} \text{ m/s}$. This means $\underline{\hspace{2cm}}$	$v = \underline{\hspace{2cm}} \text{ m/s}$. This means $\underline{\hspace{2cm}}$
The two velocities have the same $\underline{\hspace{2cm}}$ but opposite $\underline{\hspace{2cm}}$. On your v–t sketch for Graph 3, what happens at exactly $t = 5 \text{ s}$? $\underline{\hspace{4cm}}$		

TONIGHT, ONLINE

1 • Same idea, new object. You will plot 2026-XJ's position–time data. Read what it says: constant velocity, or not? What would prove it wrong?

2 • Same math. Radar: 18,000 km at 30 min, 36,000 km at 60 min. Slope → velocity in **km/h**. Watch the units.

3 • Bring one graph from today. The online exit ticket asks: what does it say, and what walk would prove it wrong. You already wrote both in Part A.

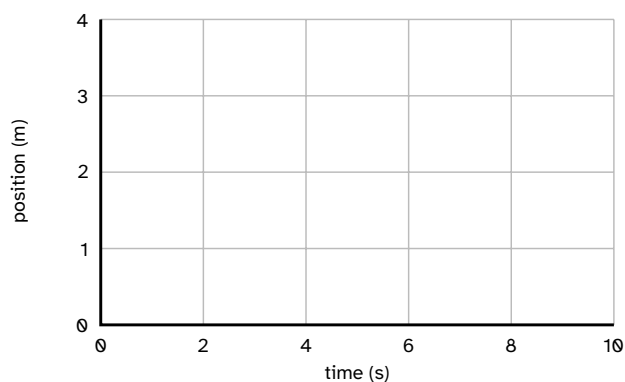
Part C • Reverse it

A partner writes a walk in words — at least one change of speed or direction. You sketch the p-t graph it should make **before** anyone moves. Then the partner walks it.

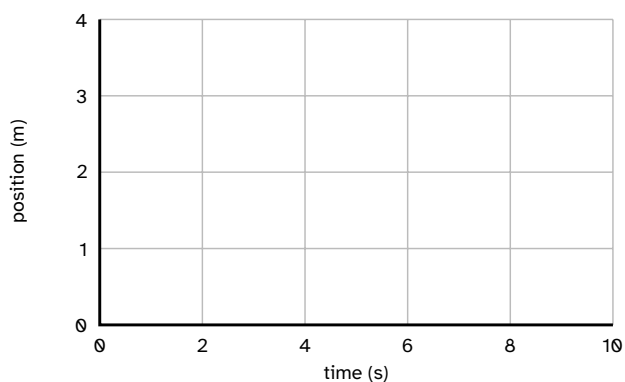
PARTNER'S WALK • WRITTEN BY: _____

Write the walk here: where it starts, each segment's direction, speed, and how long.

My sketch — BEFORE the walk



What the screen showed



Where did my sketch and the screen disagree, and whose claim was wrong — the words, my sketch, or the walk? _____

EXIT TICKET — DAY 3

A position-time graph is **flat** from 4 s to 7 s. A classmate says, "Flat means constant velocity, so the walker kept a steady speed." Is that a correct reading of the claim? Use **slope** in your answer, and sketch the matching v-t segment.

Today's Target — Rate yourself:

1 = Not Yet (I can't do this without help) · **2 = Almost** (I can do this with some help or some errors) · **3 = Got It** (I can do this independently and explain it to someone else)

Target	1	2	3	Help?
I can capture a position–time graph with a Vernier motion detector and explain what its slope tells me about velocity.	☐	☐	☐	☐